Arun Malik

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Principal Software Engineer | AI Agent Systems & Distributed Infrastructure

Redmond, WA | Open to relocation

LinkedIn | GitHub | Google Scholar | ORCID | arXiv | blog.arunmalik.dev

PROFESSIONAL SUMMARY

Principal engineer and architect who turns ambiguous, high-stakes infrastructure problems into production systems that run at **hyperscale** and that teams trust. Brings deep roots in **network security, zero-trust architecture and distributed systems**, and now applies them to **agentic AI**: autonomous agents that take real action on critical cloud infrastructure, with the guardrails, human-in-the-loop oversight and accountability needed to run them safely in production. Known for taking ideas from prototype to production as a hands-on founder and architect, setting technical strategy across 25+ organizations, and pairing shipped impact with published research, patents and conference talks.

WORK EXPERIENCE

Jul 2015 – Present

Microsoft | Software Engineer to Principal Software Engineer

Azure Networking | Viki: AI for Production Network Operations

2025 – Present

10K+ incidents/month | **99%** faster resolution | **70%+** lower LLM cost | **200+** playbooks | **0** unauthorized writes | **13** environments

- **Founded and architected Viki, an agentic AI system** that investigates and resolves Azure network incidents, taking it from hackathon idea to production as chief architect and hands-on lead; led and mentored a 10+ engineer virtual team.
- **Scaled Viki to 10K+ incidents per month (13.8K peak),** resolving automated incidents 99% faster than the overall queue (P75 ~2 hours vs. 11 days) and saving an estimated 4,000+ on-call engineer hours (~$400K) a year.
- **Extended Viki into a conversational AIOps copilot** in Teams and web chat (2,100+ users in 60 days), real-time AI validation of datacenter technician work (900+ checks in three months) and first-line support for 7+ internal network platforms.
- **Democratized AIOps automation for citizen developers** with low-code/no-code Playbook Studio and an interactive debugger: 10+ teams built 200+ playbooks with human-in-the-loop guardrails, test cycles fell from 5 days to hours, and a playbook-author agent now writes 100% of new playbooks.
- **Secured production agentic AI with zero unauthorized writes over eight months** through a zero-trust architecture in which no AI agent can reach a network device directly: granular RBAC for MCP tools (on-behalf-of identity, per-parameter least privilege), read-only by default (only ~15 of 320+ tools can write, all behind human-in-the-loop approval gates), prompt-injection scanning, JIT/PIM elevation and chain-of-custody audit; blocked 100% of 47 privilege-escalation attempts and cut policy change time 85% with policy-as-code.
- **Cut per-incident LLM cost by 70%+** while incident volume doubled by inventing Progressive Crystallization, which promotes validated agent behavior into zero-token deterministic workflows (45% of executions), under $1 per incident.
- **Built a high-velocity engineering system for production AI agents,** shipping daily with under-a-day change lead time: nightly releases across 13 environments including air-gapped Secret and Top Secret clouds, 130+ merged PRs per month, an LLM-as-judge evaluation gate and an agentic spec-to-production harness.

Azure Networking | Universal Network Inventory (UNI)

2024 – 2025

~98.5% smaller attack surface | **2.4x** scanning coverage | **580K+** owner conflicts resolved | **25+** organizations

- **Led a security initiative across 25+ organizations** to eliminate unmanaged network assets in Azure datacenters, building a CIS Control 1-style hardware asset inventory as the foundation for vulnerability management.
- **Shrank the unmanaged-device attack surface by ~98.5%** (165K+ to ~2,400) through discovery scanning, automated onboarding and staged retirement of orphaned assets; expanded vulnerability-scanning coverage 2.4x (500K to 1.2M devices) and resolved 580K+ ownership conflicts.
- **Architected secure self-service ownership APIs** with OAuth2, RBAC, audit history and credential rotation, making device-to-service ownership a single source of truth for security workflows.

Azure Cloud Security

2021 – 2024

785K subscriptions | **2.5M** virtual networks | **2M** events/hour | **96%** faster onboarding | **16,000+** services | **4x** adoption

- **Contained lateral-movement blast radius for Azure first-party services** by architecting non-overridable network micro-segmentation baselines across 785K subscriptions and 2.5M virtual networks, monitored at 2M events per hour.

- **Scaled adoption 4x (~500 to ~2,000 network managers)** after driving the move to a centrally managed successor on Azure Virtual Network Manager, architecting its safe-deployment pipeline for 20 parallel services.
- **Cut service onboarding time by 96%** (about 4 weeks to 1 day) and improved compliance across 16,000+ services through Secure by Default and network-isolation delivery.
- **Served as acting technical lead** and led identity proofs of concept and pre-rollout validation: service principals, compound identity and on-behalf-of token flows for control-plane access, plus a production-like test environment covering 105 test combinations.

Microsoft 365 | Outlook / Exchange 2015 – 2021

2B Outlook.com mailboxes consolidated | **450M** Exchange mailboxes | **Cross-country** tenant migration for data residency

- **Built the tenant-migration orchestration platform for Microsoft 365 data residency,** moving entire customer tenants across countries with identity, mail, Teams and SharePoint migrated end to end.
- **Enabled consolidation of Outlook.com (2B mailboxes) onto Exchange (450M mailboxes)** through cross-forest live mailbox migration and geo-aware placement that moved mailboxes closer to users.

Earlier Roles 2008 – 2013

L&T Infotech, Senior Software Engineer (2012–13) | TCS, Software Engineer (2010–12) | NSE.IT, Developer (2008–10)

RESEARCH, PUBLICATIONS & SPEAKING

Publications, Preprints & Writing (2026)

- Trustworthy Autonomy at Hyperscale: Granular Authorization, Human Oversight, and Progressive Crystallization for Production AI Agents. ACM HCOMP / CI 2026, accepted poster paper. [Paper]
- Progressive Crystallization: Turning Agent Exploration into Deterministic, Lower-Cost Workflows in Production. [arXiv]
- Decentralized Granular Access Control for Agentic AI Systems in Critical Infrastructure. [arXiv]
- Autonomous Incident Resolution at Hyperscale: An Agentic AI Architecture for Network Operations. [arXiv]
- From Reactive to Autonomous: Evolution of AI Operations in Cloud Network Infrastructure. [arXiv]
- MCP vs CLI for AI Agents: A Comparative Analysis of Tool Integration Strategies for Autonomous Systems. [arXiv]
- Copilot to autopilot: Why no single safeguard is enough. Microsoft Command Line. [Article]

Patent Applications

- Closed-Loop Adaptive Network Asset Inventory Management System. U.S. application filed, 2026.
- Physical Network Wiring Investigation for Accelerated Supercomputer Network Builds. U.S. application filed, 2026.

Conference Speaking, Reviewing & Judging

- **MLOps World 2026:** Speaker. Letting AI Agents Run Incident Response on a 12M-Device Network. [Speakers]
- **GlobalAI 2026:** Speaker. Progressive Crystallization in Production. [Event]
- **CNSM 2026 and NetAI 2026:** Technical Program Committee member and paper reviewer.
- **Awards judge:** Globee (Business, Leadership) and Claro, 30 evaluations; CES Innovation Awards 2027, 20 products. [Judges]

SKILLS

AI & Agents: LLMs, prompt engineering, RAG, agent orchestration, MCP, LangChain / LangGraph, Semantic Kernel, Azure OpenAI, LLM-as-judge evaluation, prompt-injection defense.

Cloud & Code: AKS / Kubernetes, Docker, Container Apps, Azure Functions, microservices, Event Hubs, Service Bus, Cosmos DB, Azure Data Explorer, Azure Storage, OpenTelemetry, Bicep; C#, .NET, Python, JavaScript, PowerShell, SQL/KQL.

Networking & Security: Azure Virtual Network Manager (AVNM), IPAM, NAT Gateway, Application Gateway, WAF v2, Private Link, NSGs, micro-segmentation, Entra ID, managed identity, RBAC, Key Vault, zero trust, STRIDE, SDL.

ACADEMIC

Master's in Software Engineering, San Jose State University 2013 – 2015

Bachelor's in Computer Engineering, University of Mumbai 2004 – 2008